

Attachment optimism, measured: a frozen world model cannot imagine grasp failures

OGBench `cube-single` · v2WM (ViT-tiny LeWM) · 2026-07-18

Probe. A frozen latent world model (WM) with encoder $E: \mathcal{O} \rightarrow \mathcal{Z}$ and forward model ϕ predicts future latents $\hat{z}_t = \phi_t(z_0, A)$ from an initial latent $z_0 = E(o_0)$ under an action sequence A . We fit a linear readout $R: \mathcal{Z} \rightarrow \mathbb{R}$ from latent to privileged block height by least squares on the expert latent cache; on held-out expert latents $R^2 = 0.990$, so R recovers block height essentially exactly. We then take action sequences A that are known—from physics-verified real execution—to produce a grasp *and miss* (the block is never lifted, real rise < 0.03 m), roll the frozen WM forward under them (three real context frames, zero action history, autoregressive frameskip-5 blocks), and read the predicted block height $R(\hat{z}_t)$. With $z_t = E(o_t)$ the real latents, define the lift $\ell(z_{0:T}) = \max_t R(z_t) - R(z_0)$ (metres). As calibration we apply the same rollout to real expert successes.

	n	real lift ℓ	WM-predicted lift $\hat{\ell}$
expert success	60	0.288	0.272 (tracks reality)
grasp-miss	112	0.002	0.210 (hallucinated grasp)

On real successes the WM predicts the lift correctly (0.272 vs. 0.288), confirming that R and ϕ are faithful. On the grasp-*and-miss* sequences the block never leaves the table (0.002), yet the WM predicts a near-full lift (0.210), above the 0.045 m grasp threshold on 100% of episodes. The predicted trajectory of a miss is thus indistinguishable, in the task-relevant coordinate, from that of a success: $R(\hat{z}_T)$ lands in the “lifted” region whether or not A really grasps. (A faint residual, 0.210 vs. 0.272, exists but sits far above threshold.) This is *attachment optimism*, and the effect is not marginal but categorical: for the exact action sequences that miss in reality, the world model imagines a successful grasp.